Program 11. Analysis of HR Dataset

i) Create KPI to show employee count, attrition count, attrition rate, attrition count, active employees, and average age.
ii) Create a Lollipop Chart to show the attrition rate based on gender category.
iii) Create a pie chart to show the attrition percentage based on Department Category- Drag department into colours and change automatic to pie. Entire view, Drag attrition count to angle. Label attrition count, change to percent, add total also, edit label.
iv) Create a bar chart to display the number of employees by Age group,
v) Create a highlight table to show the Job Satisfaction Rating for each job role based on employee count.
vi) Create a horizontal bar chart to show the attrition count for each Education field Education field wise attrition – drag education field to rows, sum attrition count to col,
vii) Create multiple donut chart to show the Attrition Rate by Gender for different Age group.

SOLUTION:

i) Create KPI to show employee count, attrition count, attrition rate, attrition count, active employees, and average age.

Create Basic Calculated Fields

1. Go to **Analysis menu → Create Calculated Field**
2. Create the following:

A) **Employee Count**

COUNT([Employee ID])

B) **Attrition Count**

IF [Attrition] = "Yes" THEN 1 ELSE 0 END

C) **Attrition Rate**

SUM([Attrition Count]) / COUNT([Employee ID])

D) **Active Employees**

COUNT([Employee ID]) - SUM([Attrition Count])

E) **Average Age** AVG([Age])

ii) Create a Lollipop Chart to show the attrition rate based on gender category.

Lollipop Chart – Attrition Rate by Gender

Prep: You already have these calculated fields
• Attrition Count
• Attrition Rate

Steps

1. Drag **Gender** → Columns
2. Drag **Attrition Rate** → Rows
3. Duplicate **Attrition Rate** on Rows (Ctrl+drag)
4. On first Marks card → Change to **Bar** (make bar very thin)
5. On second Marks card → Change to **Circle** (increase size)
6. Right-click → **Synchronize Axis**
7. Remove gridlines + show labels only on circles

This creates the stick (bar) + candy (circle).

iii) Create a pie chart to show the attrition percentage based on Department Category- Drag department into colours and change automatic to pie. Entire view, Drag attrition count to angle. Label attrition count, change to percent, add total also, edit label.

Steps: Build the Pie Chart

1. Create a **New Sheet**
2. Drag **Department** → **Color** on Marks card
3. Drag **Attrition Count** → **Angle**
4. Change **Marks type** → **Pie**
5. On the same Marks card:
    - Drag **Attrition Count** → **Label**

Steps: Show Percentage

1. Click on **Attrition Count** in Label → **Quick Table Calculation** → **Percent of Total**
2. Format Label → Show as **Percentage**

iv) Create a bar chart to display the number of employees by Age group

Step 1:

Create Age Group Calculated Field

Analysis → Create Calculated Field
Name: **Age Group**

IF [Age] < 25 THEN "<25"

ELSEIF [Age] < 35 THEN "25-34"

ELSEIF [Age] < 45 THEN "35-44"

ELSEIF [Age] < 55 THEN "45-54"

ELSE "55+"

END

This will categorize employees into age ranges.

Step 2: Build the Bar Chart

1. New Worksheet
2. Drag **Age Group → Columns**
3. Drag **Employee ID → Rows**
4. Right-click **Employee ID** → Convert to **Count**
5. Click **Sort → Descending by Count(Employee ID)**

The tallest bar = most employees in that age category.

Step 3: Improve Formatting

• Change Marks type to **Bar** if not set
• Add **Count(Employee ID) → Label**
• Drag **Age Group → Color** (optional)

Remove Gridlines

1. Go to the **Format** menu at the top
2. Select **Lines**
3. In the panel on the left:
   • Under **Grid Lines**, set **Rows** → *None*
   • Set **Columns** → *None*

v) Create a highlight table to show the Job Satisfaction Rating for each job role based on employee count.

Steps

- Drag **Job Role → Rows**
- Drag **Job Satisfaction → Columns**
- Drag **Employee Count** (or Number of Records) → **Color**
  It will start looking like a heatmap.
- On Marks card → Change **Automatic** to **Bar**
  Cells will become blocks that fill space.
- Drag **Employee Count** again → **Label**
  Now you can see the values in each color bar.
- Adjust color palette
  • Color → choose a sequential palette like **Orange–Blue** or **Red–Green**
  • Increase intensity if needed

vi) Create a horizontal bar chart to show the attrition count for each Education field Education field wise attrition – drag education field to rows, sum attrition count to col,

Steps:

- Drag **Education Field → Rows**
  This sets different education groups on the vertical axis.
- Drag **Attrition Count → Columns**
  Bars will automatically appear.
- On the Marks card, ensure **Bar** is selected
  (This should be automatic, but always good to confirm.)
- Sort by attrition count
  • Click the **Sort** icon on toolbar (largest to smallest)
  ✅ Gives a better visual order
- Add labels
  • Drag **Attrition Count** → Label

vii) Create multiple donut chart to show the Attrition Rate by Gender for different Age group.

Donuts can be a little stubborn in Tableau
Steps:

- Drag **Age Group → Columns**
  Creates one donut per age group.
- Drag **Gender → Color**
  So each donut is split into Male vs Female.
- Drag **Attrition Rate → Angle**
  Slices show percentage.
- Convert to **Pie Chart**
  • Marks Card → Change to **Pie**
  • Size → Increase so pies are larger
- Duplicate the measure to build donut (the secret trick 😎)
  • Hold Ctrl + drag **Attrition Rate** again to **Rows**

You will now see two pie charts stacked.